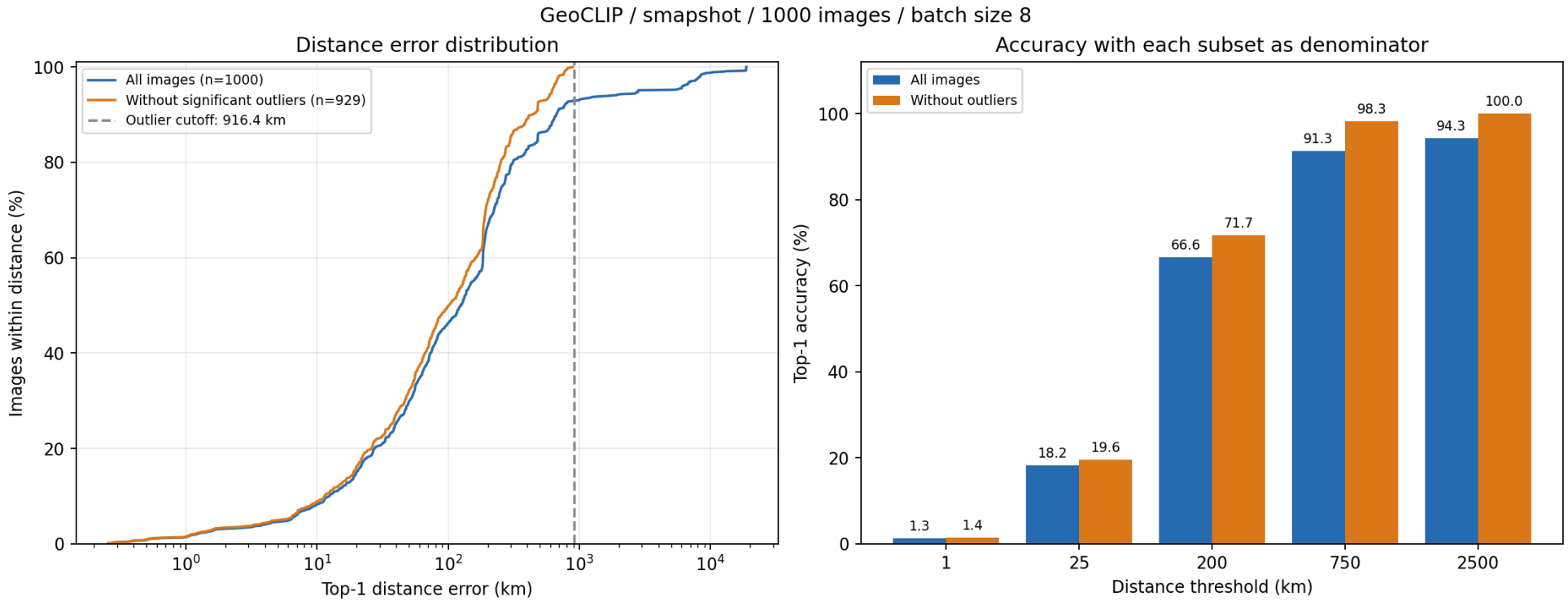
Benchmark: GeoCLIP (test dataset: 1000 pictures from Smapshot)




Median error: 125.05 km  Average error: 671.02km

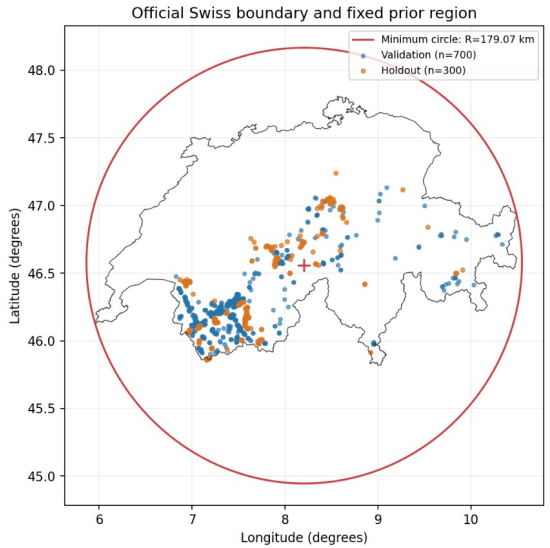
The average is significantly boosted by a few extreme values greater then 2500km (~5.7% of data points). A visualization of results cropping extreme values can better demonstrate the result. Cropping criterion: error > 916.43km (7.1%)

Median error: 101.42km  Average error:166.34 km

‘Benchmark’: GeoCLIP with output limited in Switzerland

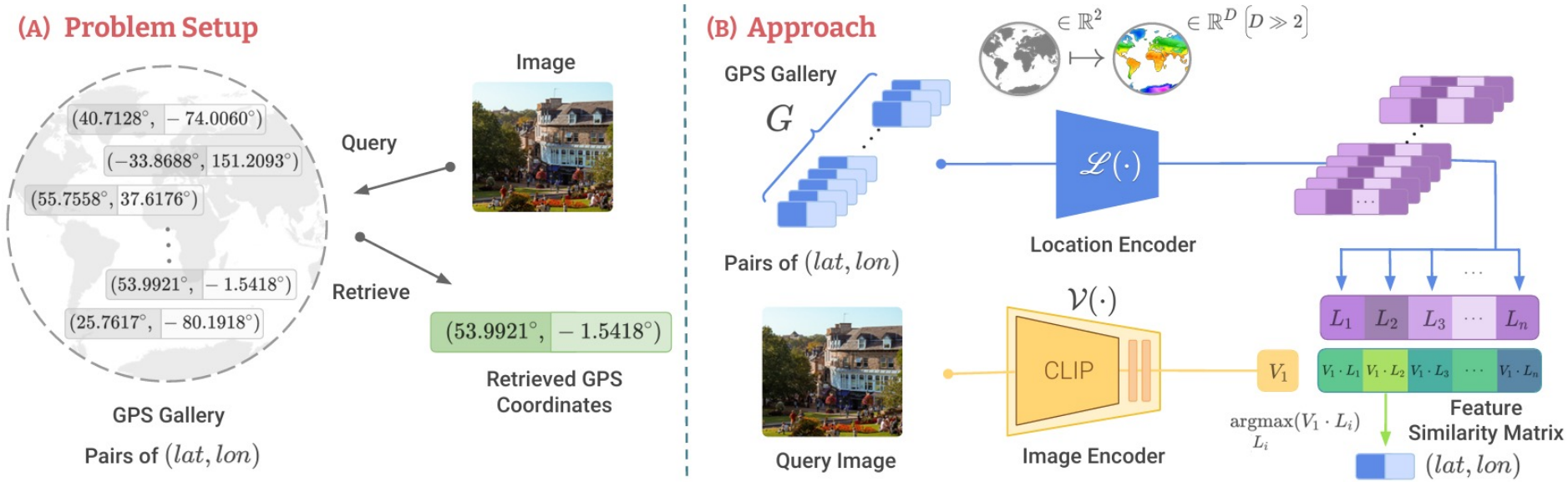
$$U(g) = \left[\max\left(0, \frac{d(g,c) - R}{b}\right)\right]^2$$

$$s_{\text{regional}}(I, g) = s_{\text{GeoCLIP}}(I, g) - \lambda U(g)$$




| Indicator / 1000 pics | Original Model | Swiss Output Model |
|---|---|---|
| Median | 125.05 | 69.04 |
| Average | 665.83 | 92.37 |

Identify factors



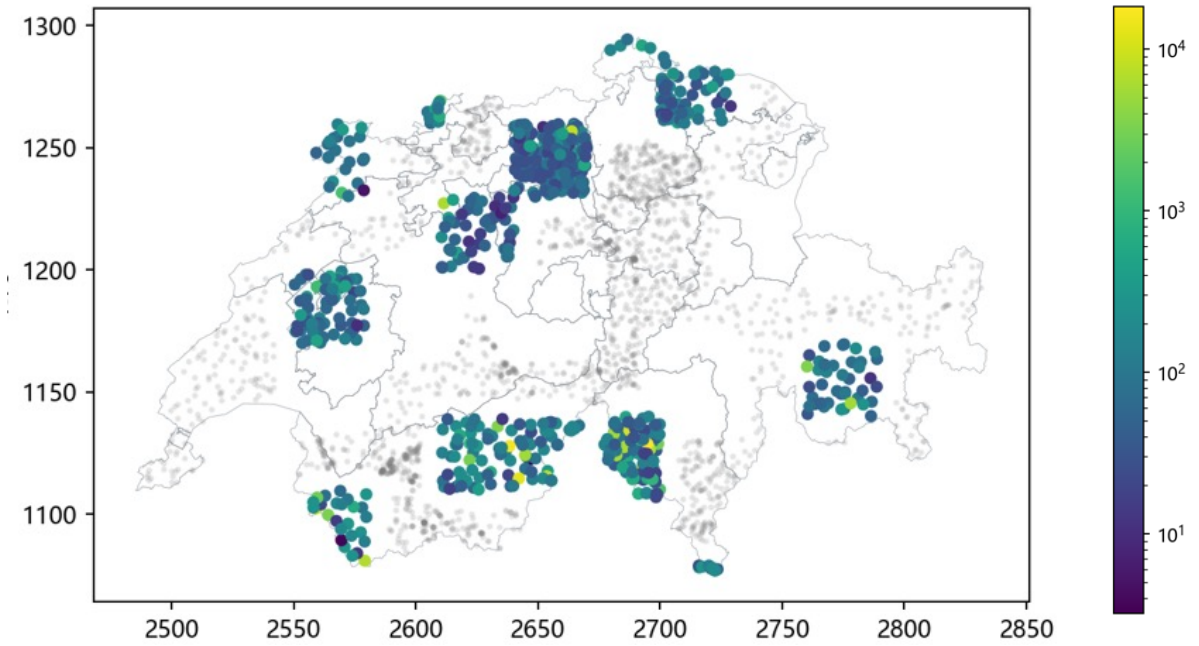

Bottleneck of classification accuracy: limited accuracy in image feature extraction or limited (latitude, longitude) pairs?

Identify factors : Naïve training

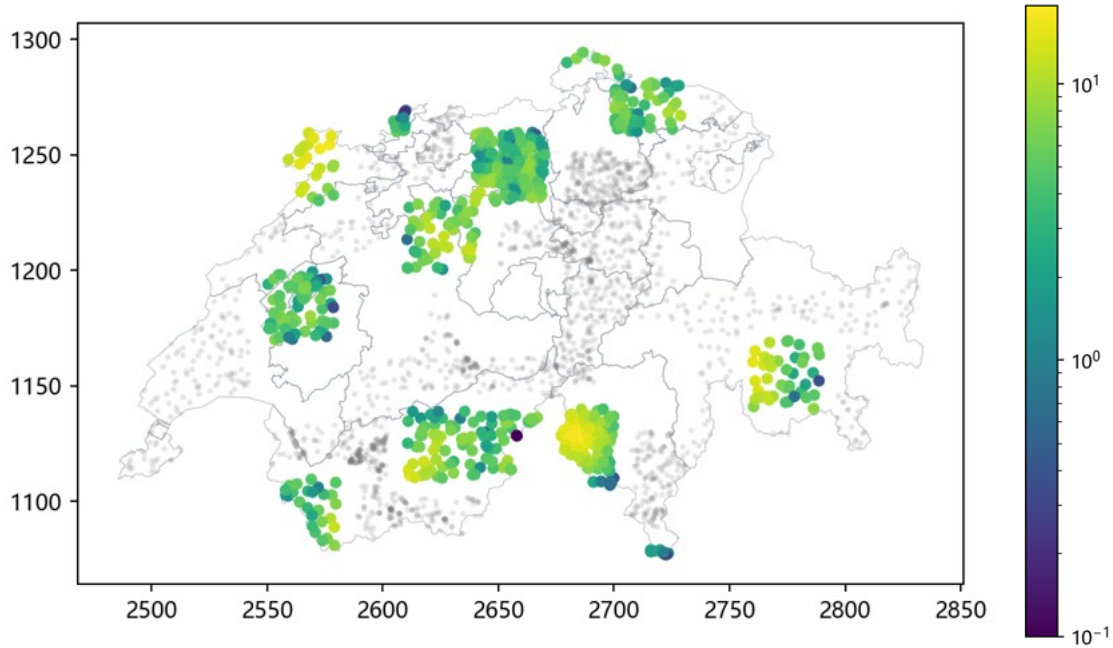
Unfreeze MLP & location encoder: train 550/validate 150/test 300

| Indicator | Median Output | Average Output | 25km Accuracy |
|---|---|---|---|
| Original model | 79.02 | 754.44 | 28.00% |
| MLP finetune | 27.74 | 378.79 | 45.67% |
| MLP + location encoder finetune | 25.02 | 306.63 | 50.00% |

Identify factors : Spatial distribution




Spatial distribution of training errors




Spatial distribution of closest possible output